# JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

# Design and Implementation of an Intelligent ISP Billing and Management System Using Multiple Linear Regression for Revenue Prediction

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**Reg No: SCT221-0271/2022**

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*A proposal/research project submitted to the Department of Information Technology in the School of Computing and Information Technology in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Information Technology, Jomo Kenyatta University of Agriculture and Technology.*

**2025/2026**

## DECLARATION

This proposal is my original work and has not been presented for a degree in any other University.

Signature: ………………………………………..

Date: ………………………………………..

**WARREN CHRIS**

This proposal has been submitted for examination with my approval as University Supervisor.

Signature: ………………………………………..

Date: ………………………………………..

**MR. JOMO NJENGA**

**ACKNOWLEDGEMENT**

I would like to express my sincere gratitude to my supervisor, Mr. Jomo Njenga, for his invaluable guidance, constructive criticism, and continuous support throughout the development of this project proposal. His insights and encouragement greatly contributed to shaping the direction and quality of this research.

I also extend my appreciation to the lecturers and staff of the Jomo Kenyatta University of Agriculture and Technology, Department of Information Technology, for equipping me with the knowledge and technical skills necessary to undertake this project.

Special thanks to my family and friends for their unwavering support, motivation, and encouragement throughout my academic journey.

Finally, I am grateful to everyone who contributed directly or indirectly to the successful completion of this proposal.

**ABSTRACT**

The Internet Service Provider (ISP) industry in Kenya continues to experience rapid growth due to increased demand for affordable and reliable internet connectivity. Despite this growth, many small to medium-sized ISPs still rely on manual or semi-automated tools such as spreadsheets to manage customer records, billing cycles, and payments. These approaches result in revenue leakage, billing inaccuracies, delayed service provisioning, and poor customer experience (Khan, Ali, & Khan, 2022).

This research proposes the development of an ISP Billing and Management System that leverages modern web technologies and intelligent automation to address these challenges. The proposed system will automate customer registration, subscription management, invoice generation, and M-Pesa payment reconciliation (Safaricom PLC, 2024). Additionally, the system integrates mathematical models for prediction and anomaly detection, while a LLaMA-based large language model (LLM) is used for interpretation, explanation, and decision support. This hybrid approach ensures accuracy, transparency, and scalability without training models from scratch. The Waterfall model to be used follows a structured and sequential approach consisting of requirement analysis, system design, implementation, testing, deployment, and maintenance (Sommerville, 2016). The expected outcome is a secure, scalable, and efficient system that enhances operational efficiency for ISPs while improving transparency and convenience for customers.

In addition, the system incorporates an integrated support ticketing module that enables customers to submit service-related issues such as connectivity problems and billing disputes, which are tracked, assigned, and resolved by administrators and support staff through a centralized interface.

## TABLE OF CONTENTS

**LIST OF TABLES**



**LIST OF FIGURES**

## ACRONYMS

ISP – Internet Service Provider

API – Application Programming Interface

AI – Artificial Intelligence

JWT – JSON Web Token

LLM – Large Language Model

NLP – Natural Language Processing

ORM – Object Relational Mapping

STK – SIM Toolkit

UAT – User Acceptance Testing

## DEFINITION OF TERMS

**Billing Cycle:** A fixed period during which internet usage is measured and billed.

**STK Push:** A payment request initiated to a customer's mobile phone via M-Pesa.

**LLM:** Large Language Model used for natural language processing tasks.

# CHAPTER 1: INTRODUCTION

## 1.1 Background

Globally, Internet Service Providers have transitioned from manual billing systems to fully automated, intelligent platforms that integrate payments, customer management, and analytics. In developed markets, ISPs leverage real-time billing engines, self-service portals, and AI-powered customer support to improve efficiency and service quality. These systems reduce revenue loss, enhance transparency, and enable data-driven decision-making (Kshetri, 2021).

In Kenya, however, many small to medium-sized ISPs still depend on spreadsheets and manual reconciliation of payments, particularly for mobile money transactions. With the dominance of M-Pesa as the primary payment method, manual matching of transaction codes to customers introduces delays and errors (Gikandi & Bloor, 2021). As customer bases expand, these inefficiencies significantly affect revenue collection and customer satisfaction.

## 1.2 Project Overview

This research focuses on the design and development of an ISP Billing and Management System that integrates subscription management, automated billing, M-Pesa payment processing, and intelligent analytics. Artificial intelligence is implemented using a hybrid approach where mathematical models handle prediction and anomaly detection, while LLMs support interpretation and decision-making. The project is grounded in modern computational principles such as web-based distributed systems, RESTful APIs, secure authentication mechanisms, and artificial intelligence for natural language processing.

The system aims to provide a centralized platform for ISPs to manage customers, service plans, payments, and support operations while offering customers a self-service portal for account monitoring and payments.

## 1.3 Statement of the Problem

Despite advancements in payment and networking technologies, many ISPs in Kenya experience revenue leakage due to untracked subscription expiries and delayed payment reconciliation (Gikandi & Bloor, 2021). Manual processes increase the likelihood of human error, resulting in billing disputes and service interruptions. Customers lack access to real-time usage data and invoices, forcing them to rely on support staff for basic account information.

These challenges limit scalability, reduce operational efficiency, and negatively impact customer experience. The problem is further compounded by the absence of intelligent systems capable of analyzing usage trends and predicting customer churn using mathematical models combined with LLM-based hybrid approaches (Xu et al., 2020).

## 1.4 Proposed Solution

This research proposes the development of an intelligent ISP Billing and Management System that automates subscription tracking, billing, and payment reconciliation. The system incorporates AI-driven customer support powered by hybrid intelligence (mathematical models + LLaMA-based LLMs) to enhance service delivery and decision-making (Laranjo et al., 2018). The proposed solution leverages modern web technologies, integrates seamlessly with M-Pesa, and adopts contemporary architectural best practices to ensure scalability and security.

## 1.5 Objectives

**General Objective**

To design and implement an intelligent ISP Billing and Management System that automates billing, payment processing, and customer management.

**Specific Objectives**

1. To design and document a secure ISP billing system architecture, including at least three core modules (customer registration, subscription management, and role-based access control), implemented using RESTful APIs and JWT authentication, and verified against basic web security standards (authentication, authorization, and data encryption) through a design review checklist, by the end of Phase 2 (Week 6 of the project timeline).
2. To implement an automated billing and invoicing module that generates monthly invoices and processes customer payments through M-Pesa STK Push that integrates directly with the Safaricom M-Pesa Daraja API, achieving successful transaction

recording and invoice updates for at least 95% of test payment scenarios by the system implementation phase (Safaricom PLC, 2024).

3. To develop a hybrid AI-powered customer support and analytics module combining mathematical models (for prediction and anomaly detection) with NLP and LLMs for interpretation and explanation, achieving at least 70% response accuracy (Laranjo et al., 2018; Xu et al., 2020).
4. To test and evaluate the developed system by conducting performance, security, and usability tests, ensuring the system handles at least 100 concurrent users, passes basic security checks through unit testing, integration testing, and user acceptance testing (UAT), and attains a minimum 70% usability satisfaction score from test users prior to project submission.
5. To design and implement a support ticketing and issue management module that enables customers to submit service-related complaints and allows administrators to track, assign, prioritize, and resolve tickets efficiently.

## 1.6 Research Questions

1. How can intelligent automation improve billing accuracy and payment reconciliation for ISPs?
2. What system architecture best supports scalable ISP billing and management operations?
3. How can AI techniques enhance customer support and administrative decision-making?
4. How effective is the proposed system in reducing revenue leakage and operational overhead?

## 1.7 Justification

This research is motivated by the increasing need for efficient ISP management solutions in Kenya. The proposed system addresses current operational gaps by introducing automation and intelligence into billing and customer management. The research contributes to the application of

AI and modern web technologies in the ISP domain and provides a practical solution aligned with current technological trends.

## 1.8 Proposed System Methodology

The project will adopt the Waterfall software development model as the primary development methodology. The Waterfall model follows a structured and sequential approach consisting of requirement analysis, system design, implementation, testing, deployment, and maintenance (Sommerville, 2016).

## 1.9 Scope

The study focuses on small to medium-sized ISPs operating in Kenya. The system addresses customer management, billing, payments, and support operations. The research is limited to software-based solutions and does not cover physical network infrastructure deployment.

## CHAPTER 2: LITERATURE REVIEW

### 2.1 Introduction

This chapter reviews existing literature related to ISP billing systems, payment integration, and artificial intelligence applications in service management. The review establishes a theoretical foundation and identifies gaps addressed by this research.

### 2.2 Theoretical Review

This research is grounded in several theoretical concepts related to information systems, billing automation, payment integration, and artificial intelligence. Understanding these theories provides a foundation for designing an intelligent ISP Billing and Management System.

One key concept is **subscription-based billing systems**, which involve charging customers recurring fees based on predefined service plans. Subscription billing theory emphasizes accuracy, timeliness, and transparency in billing cycles to prevent revenue leakage. In ISP environments, subscription billing is tightly coupled with service provisioning, where access to services must dynamically respond to payment status (Alalwan et al., 2020).

Another important concept is **mobile money integration theory**, particularly within developing economies. Mobile money systems such as M-Pesa operate through APIs that enable third-party systems to initiate, receive, and reconcile payments. The theory highlights the importance of real-time transaction validation, fault tolerance, and security in financial systems. Poor integration leads to delayed confirmations and reconciliation errors, which negatively impact service delivery.

The study also draws from **information systems automation theory**, which posits that automating repetitive and data-intensive tasks reduces human error and increases operational efficiency. Automated billing engines, notification systems, and rule-based decision engines are examples of applied automation within enterprise systems.

Artificial Intelligence (AI) theory is central to this research, particularly in the areas of **rule-based systems** and hybrid approaches combining mathematical models with **Large Language Models (LLMs)**. Rule-based systems operate on predefined logical conditions (e.g., if payment is overdue, suspend service) and are valued for their predictability and transparency. However, they lack adaptability and natural interaction capabilities.

LLMs, on the other hand, are generative AI models trained on large datasets to understand and produce human-like language. In information systems, LLMs are increasingly applied in customer support, data querying, and decision support systems. While LLMs offer flexibility and conversational interfaces, they require contextual grounding and guardrails to ensure accuracy and reliability (Kshetri, 2021).

The hybrid AI module supports the ticketing system by analyzing historical ticket data using mathematical models to identify recurring issues and abnormal patterns, while the LLaMA-based language model assists in ticket categorization, prioritization, and natural language summarization for administrative decision support.

The combination of rule-based logic, mathematical models with LLMs represents a hybrid intelligence model, where deterministic rules handle critical operations while AI models enhance interaction and insight generation. This theoretical integration informs the design of the proposed system.

## 2.3 Case Study Review

Several existing systems and studies provide insight into ISP billing automation and intelligent service management.

A study on automated billing systems in small telecommunications firms demonstrated that real-time subscription tracking significantly reduced revenue leakage and billing disputes (Khan, Ali, & Khan, 2022). The study highlighted that automation alone was insufficient without proper integration with payment platforms, especially in mobile-money-dominated markets.

In East Africa, Safaricom’s M-Pesa ecosystem has enabled the development of numerous third-party billing and payment platforms. Case studies on M-Pesa-integrated utility billing systems reveal improved payment turnaround times and higher customer satisfaction (Gikandi & Bloor, 2021; Safaricom PLC, 2024). However, many systems lack advanced analytics and customer-facing intelligence.

Globally, ISP management platforms such as WHMCS and Splynx provide comprehensive billing and customer management solutions. These systems offer automated invoicing and payment gateways but are often costly and not fully optimized for local payment methods such as M-Pesa. Additionally, their AI capabilities are limited or absent, relying mostly on static dashboards (Sommerville, 2016)..

Recent studies on AI-powered customer support systems indicate that chatbots and conversational agents can handle up to 60% of routine customer queries effectively (Xu, Liu, Guo, Sinha, & Akkiraju, 2020; Laranjo et al., 2018). In service industries, AI assistants reduce support workload and improve response times. However, research also notes that generic chatbots fail when not integrated with real-time system data.

These case studies demonstrate the potential of automation and AI while revealing gaps in localization, affordability, and intelligent integration for small ISPs. This research addresses these gaps by proposing a localized, AI-enhanced ISP Billing and Management System that integrates automation, mobile payments, and intelligent analytics into a unified platform using a hybrid approach (mathematical models + LLaMA-based LLMs).

## 2.4 Predictive Analytics in Telecommunications

Predictive analytics has become a critical component in modern telecommunications and subscription-based service industries. Telecommunications companies generate large volumes of customer data, including usage patterns, payment history, service subscriptions, and support interactions. Statistical and regression-based models are widely used to analyze this data to forecast revenue, predict customer behavior, and identify operational risks (Khan et al., 2022).

Multiple Linear Regression (MLR) is commonly applied in revenue forecasting within subscription systems. The model estimates the relationship between a dependent variable (e.g., monthly revenue) and multiple independent variables such as number of active subscribers, average data usage, payment delays, and service plan types. The regression equation enables organizations to quantify the impact of each variable on revenue performance and make data-driven decisions (Sommerville, 2016).

In ISP environments, predictive models can estimate expected revenue for upcoming billing cycles and detect deviations from normal patterns. When combined with automated billing systems, predictive analytics improves financial planning and reduces revenue leakage (Khan et al., 2022). Unlike black-box machine learning models, regression-based approaches are transparent and interpretable, making them suitable for academic and small-scale enterprise implementations.

The integration of predictive analytics into ISP billing platforms therefore enhances strategic planning, improves forecasting accuracy, and supports proactive decision-making.

### 2.5 Customer Churn Prediction Models

Customer churn refers to the rate at which customers discontinue their subscriptions within a given period. In telecommunications industries, churn significantly affects revenue stability and long-term profitability. Studies show that retaining existing customers is more cost-effective than acquiring new ones, making churn prediction a critical component of service management systems (Xu et al., 2020).

Logistic regression and other statistical classification models are frequently used to predict churn probability. These models analyze behavioral variables such as payment delays, frequency of support tickets, data usage patterns, and subscription duration. Customers exhibiting abnormal usage decline or repeated late payments may have a higher probability of churn (Laranjo et al., 2018; Xu et al., 2020).

In the context of ISP management systems, churn prediction models can generate risk scores for individual customers. Administrators can then implement retention strategies such as targeted notifications, promotional offers, or proactive customer support. The integration of churn prediction within billing systems transforms them from passive transaction processors into proactive revenue protection platforms.

### 2.6 Anomaly Detection in Billing Systems

Billing systems are vulnerable to operational anomalies such as incorrect invoice generation, duplicate transactions, payment mismatches, and fraudulent activities. Anomaly detection techniques use statistical thresholds and regression-based residual analysis to identify deviations from expected behavior (Khan et al., 2022).

For example, if the predicted monthly revenue based on regression analysis significantly differs from actual collected revenue, the system can flag a potential billing inconsistency. Similarly, unusual spikes in usage data may indicate technical faults or misuse (Gikandi & Bloor, 2021).

Mathematical anomaly detection models rely on calculating variance, residual errors, and statistical confidence intervals. These techniques are computationally efficient and suitable for small to medium-sized enterprise systems. When integrated into ISP billing platforms, anomaly detection enhances reliability, improves financial transparency, and strengthens internal control mechanisms.

## 2.7 Cybersecurity in Financial and Billing Systems

Billing and payment platforms handle sensitive financial and personal data, making them prime targets for cyber threats. Secure system architecture is therefore essential in ISP billing systems, particularly when integrating third-party payment APIs such as mobile money platforms (Mwangi & Kimani, 2023).

Modern web security frameworks emphasize authentication, authorization, encryption, and secure API communication. JSON Web Token (JWT) authentication mechanisms enable secure session management and role-based access control. Secure webhook validation ensures that payment callbacks from external APIs are verified before transaction confirmation (Safaricom PLC, 2024).

Additionally, encryption standards such as HTTPS and secure hashing algorithms protect data in transit and at rest. The Open Web Application Security Project (OWASP) outlines common vulnerabilities including injection attacks, cross-site scripting (XSS), and insecure direct object references (Sommerville, 2016). Incorporating these security principles during system design improves resilience and user trust.

Incorporating cybersecurity measures into ISP billing platforms ensures compliance with financial data protection standards and safeguards against revenue loss due to malicious activities.

## 2.8 Conversational AI and Large Language Models in Enterprise Systems

Conversational AI systems powered by Large Language Models (LLMs) are increasingly deployed in enterprise environments to enhance customer support and internal decision-making. Unlike traditional rule-based chatbots, LLMs are capable of understanding context, generating natural language responses, and summarizing complex information (Laranjo et al., 2018).

In billing systems, conversational AI can assist customers in understanding invoices, checking subscription status, and resolving common service inquiries. However, research highlights that LLMs must be grounded in real-time system data to prevent inaccurate or fabricated responses (Kshetri, 2021).

A hybrid approach combining deterministic system rules with LLM-based interpretation ensures reliability. Mathematical models provide structured predictions and analytics, while the LLM translates technical outputs into user-friendly explanations. This integration enhances usability while maintaining operational control.

The use of conversational AI within ISP billing systems therefore improves service accessibility, reduces support workload, and enhances overall user experience.

## 2.9 Cloud Computing and Scalable System Architecture

Cloud computing has transformed the deployment and scalability of modern information systems. Cloud-based infrastructure enables dynamic resource allocation, high availability, and cost-efficient service delivery (Sommerville, 2016).

Subscription-based platforms such as ISP billing systems require scalable architectures capable of handling concurrent users, real-time transaction processing, and data analytics. Containerization technologies and modular system designs improve maintainability and deployment consistency.

In mobile-money-integrated systems, reliable API connectivity and fault tolerance mechanisms are essential to ensure seamless transaction processing (Safaricom PLC, 2024). Secure and scalable architectures therefore form the foundation for sustainable ISP billing platforms, particularly in emerging digital economies.

## 2.10 Integration and Architecture

System integration is a critical aspect of modern information systems. This research adopts a layered architectural approach combining presentation, application, logic, and data layers.

The frontend layer provides user interaction through web interfaces built using modern JavaScript frameworks. The backend layer exposes RESTful APIs that handle business logic, authentication, billing operations, and integrations with external services such as M-Pesa.

Integration with the M-Pesa Daraja API requires secure token-based authentication, callback handling, and transaction validation mechanisms. Architectural patterns such as webhook listeners and message queues are considered to ensure reliability during high transaction volumes.

For AI integration, the architecture incorporates an AI service layer that combines mathematical models for analytics with LLM APIs. This layer retrieves contextual data such as user subscription status, usage, and payment history before generating responses. This approach ensures that AI outputs are grounded in system data rather than generic responses.

The system architecture follows the Model-View-Controller (MVC) pattern, promoting separation of concerns and maintainability. Containerization using Docker enables consistent deployment across development and production environments.

## 2.11 Summary

The literature reviewed emphasizes the importance of automation, secure payment integration, and intelligent interaction in modern ISP management systems. Subscription billing theory, mobile money integration principles, and AI frameworks collectively inform the design of the proposed solution.

## 2.12 Research Gaps

Despite the availability of billing and ISP management platforms, several gaps remain. Many existing systems are not optimized for mobile-money-first economies such as Kenya. Additionally, most solutions lack intelligent analytics and conversational interfaces tailored for small and medium-sized ISPs.

Furthermore, current AI implementations in billing systems are either overly generic or disconnected from real-time operational data. This research addresses these gaps by proposing a localized, AI-enhanced ISP Billing and Management System that integrates automation, mobile payments, and intelligent analytics into a unified platform.

Existing solutions lack localized payment optimization and intelligent analytics tailored for small ISPs. This research addresses these gaps by integrating M-Pesa and AI-driven insights.

**APPENDICES**

Appendix A: References

Ajzen, I. (2020). The theory of planned behavior: Frequently asked questions. *Human Behavior and Emerging Technologies*, 2(4), 314–324. https://doi.org/10.1002/hbe2.195

Alalwan, A. A., Dwivedi, Y. K., Rana, N. P., & Williams, M. D. (2020). Consumer adoption of mobile payment systems: A meta-analytical review. *Information Systems Frontiers*, 22(3), 555–573. https://doi.org/10.1007/s10796-018-9875-4

Gikandi, J. W., & Bloor, C. (2021). Adoption and use of mobile money services in developing countries: Evidence from Kenya. *Journal of African Business*, 22(3), 356–374. https://doi.org/10.1080/15228916.2020.1838474

Khan, S., Ali, A., & Khan, F. (2022). Design and implementation of automated billing systems for internet service providers. *International Journal of Computer Applications*, 174(22), 1–8.

Kshetri, N. (2021). The economics of mobile money in developing countries. *Telecommunications Policy*, 45(2), 102–115. https://doi.org/10.1016/j.telpol.2020.102102

Laranjo, L., Dunn, A. G., Tong, H. L., Kocaballi, A. B., Chen, J., Bashir, R., … Coiera, E. (2018). Conversational agents in healthcare: A systematic review. *Journal of the American Medical Informatics Association*, 25(9), 1248–1258. https://doi.org/10.1093/jamia/ocy072

Mwangi, I., & Kimani, S. (2023). Secure API integration strategies for mobile money platforms. *African Journal of Information Systems*, 15(2), 45–60.

Safaricom PLC. (2024). *Daraja API documentation*. Safaricom Developer Portal. https://developer.safaricom.co.ke/

Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson Education.

Xu, A., Liu, Z., Guo, Y., Sinha, V., & Akkiraju, R. (2020). A new chatbot for customer service on social media. *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 3506–3510. https://doi.org/10.1145/3025453.3025496

Appendix B: Project Schedule

The project will be conducted over a period of three months in line with the university's final year project timeline. The schedule is divided into key phases to ensure systematic progression and timely completion of the project.

**Table B.2 Project Schedule**

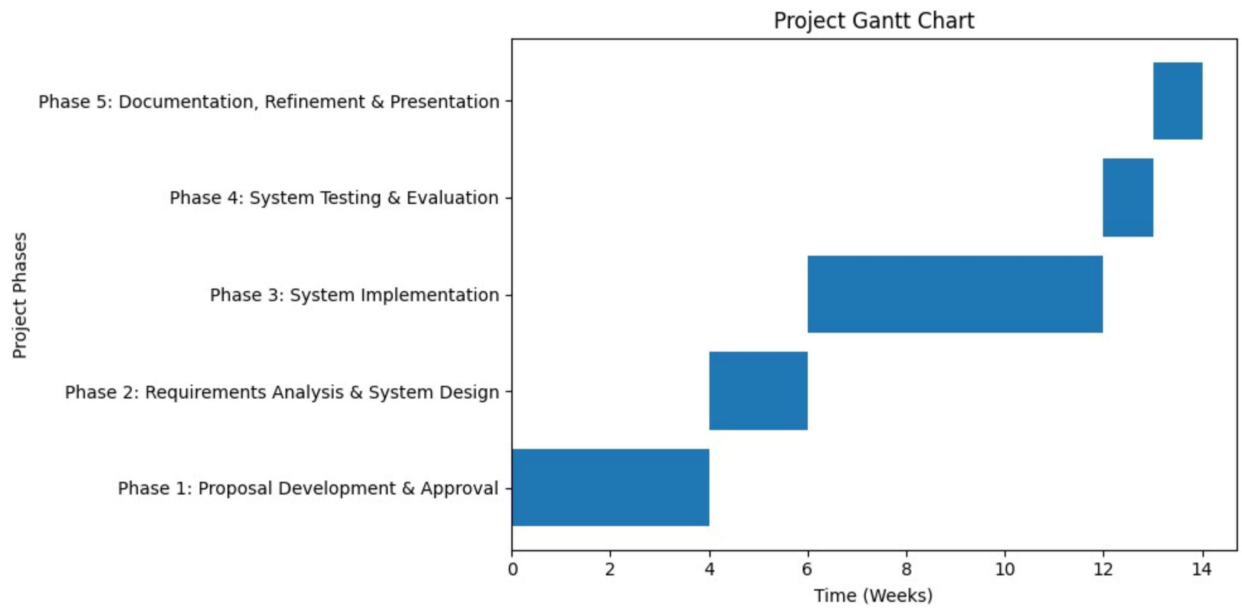


*Figure D1: The diagram illustrates a grant chart for the project schedule*

Appendix C: Project Budget

The estimated budget for the project is outlined below. The project primarily relies on open-source technologies to minimize costs.

| ITEM | DESCRIPTION | Estimated Cost |
|---|---|---|
| Development Laptop & Utilities | Pro-rated academic usage cost | 0 |
| Backup Storage | Data protection & system backups | 1000 |
| Domain Registration | System hosting domain | Ksh.1000 |
| Cloud Hosting & Testing | Backend & Database deployment | Ksh.3000 |
| AI API Usage | Token usage & testing | Ksh.2000 |
| M-Pesa Daraja Integration | SMS callbacks, testing costs | Ksh.1500 |
| Email Services | Invoice and notification delivery | Ksh.2500 |
| Performance & Load Testing | Concurrent user simulation tools | Ksh.1500 |
| Presentation Materials | Slides, demo preparation | Ksh.500 |
| Miscellaneous & Contingency | Unexpected expenses | Ksh.3000 |
| **Total** | | Ksh.16000 |

**Table B.2 Project Budget Estimate**

Appendix D: Project Resources

The successful completion of this project will require both hardware and software resources. Hardware resources include a personal computer with sufficient processing power and memory to run development tools, databases, and containerized services.

| CATEGORY | RESOURCE | PURPOSE |
|---|---|---|
| Hardware Resources | Development Laptop/PC | System development, coding, testing, and deployment |
| | External Storage / Cloud Backup | Data backup and recovery |
| | Stable Internet Connection | API integration, cloud deployment, version control |
| Software Resources | Node.js & Express.js | Backend development and API creation |
| | React.js | Frontend development (User Interface) |
| | MySQL | Database management system |
| | Docker | Containerization and deployment consistency |
| | Git & GitHub | Version control and source code management |
| | Visual Studio Code | Code editor and development environment |
| AI & Analytical Tools | Multiple Linear Regression Model | Revenue prediction and anomaly detection |
| | LLaMA-based LLM API | Natural language support and decision assistance |
| Third-Party APIs & Services | M-Pesa Daraja API | Payment processing and transaction validation |
| | Email Service (SMTP/API) | Invoice and notification delivery |
| Testing & Security Tools | Postman | API testing |
| | Load Testing Tools | Performance evaluation |
| | JWT Authentication | Secure access control |
| Human Resources | Student Researcher | System design, development, and documentation |
| | Project Supervisor | Academic guidance and evaluation |
| | Test Users | User acceptance testing |

Appendix E: System Diagrams

This appendix outlines the key system diagrams required to support the design and implementation of the proposed ISP Billing and Management System. All diagrams should be clearly labeled and captioned according to APA standards and referenced within the main report.

**Figure D.2 System Architecture Diagram**

This diagram illustrates the high-level architecture of the system, showing the interaction between the frontend (React.js), backend server (Node.js and Express), database (MySQL), third-party services (M-Pesa Daraja API, Email services), and the AI service layer. The diagram demonstrates data flow and integration points across system components.

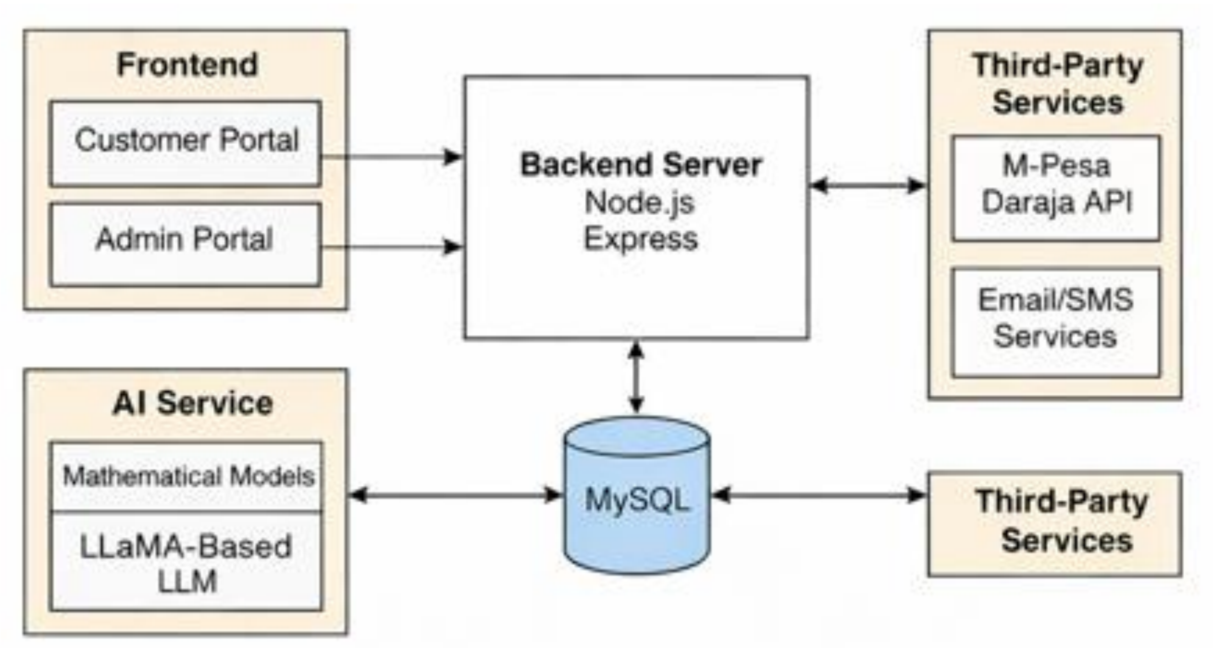


*Figure D2: The diagram illustrates the component-based architecture of the ISP billing and management system.*

**Figure D.3 Use Case Diagram**

The use case diagram represents the functional interactions between system actors and the system. The primary actors include Administrator, Support Staff, and Customer. Key use cases include user registration, subscription management, invoice generation, payment processing, viewing usage statistics, and AI-assisted support.

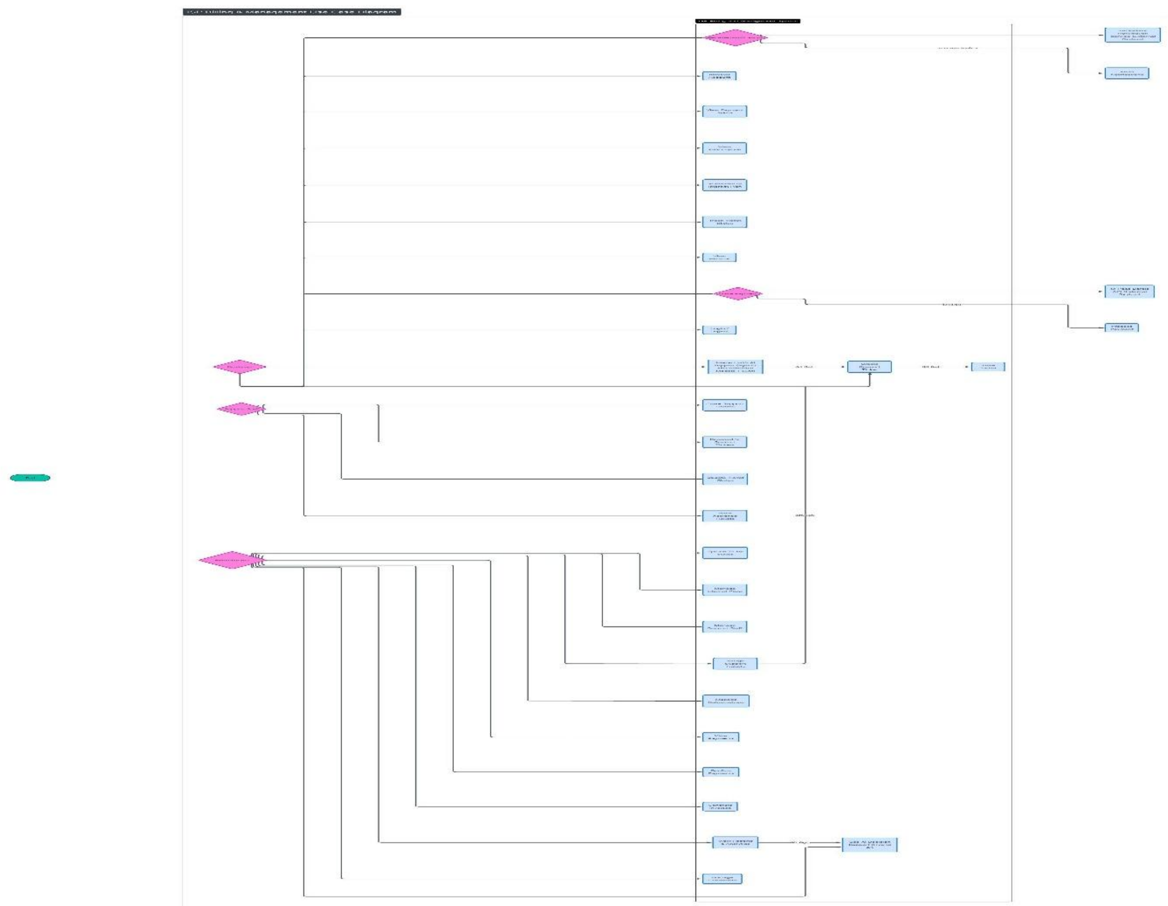

*Figure D3: The diagram illustrates the USE-CASE diagram of the ISP billing and management system.*

**Figure D.4 Entity Relationship Diagram (ERD)**

The ERD models the database structure of the system, highlighting key entities such as Users, Roles, Subscriptions, Service Plans, Invoices, Payments, and Support Tickets. Relationships between entities are defined to illustrate data integrity and system logic.

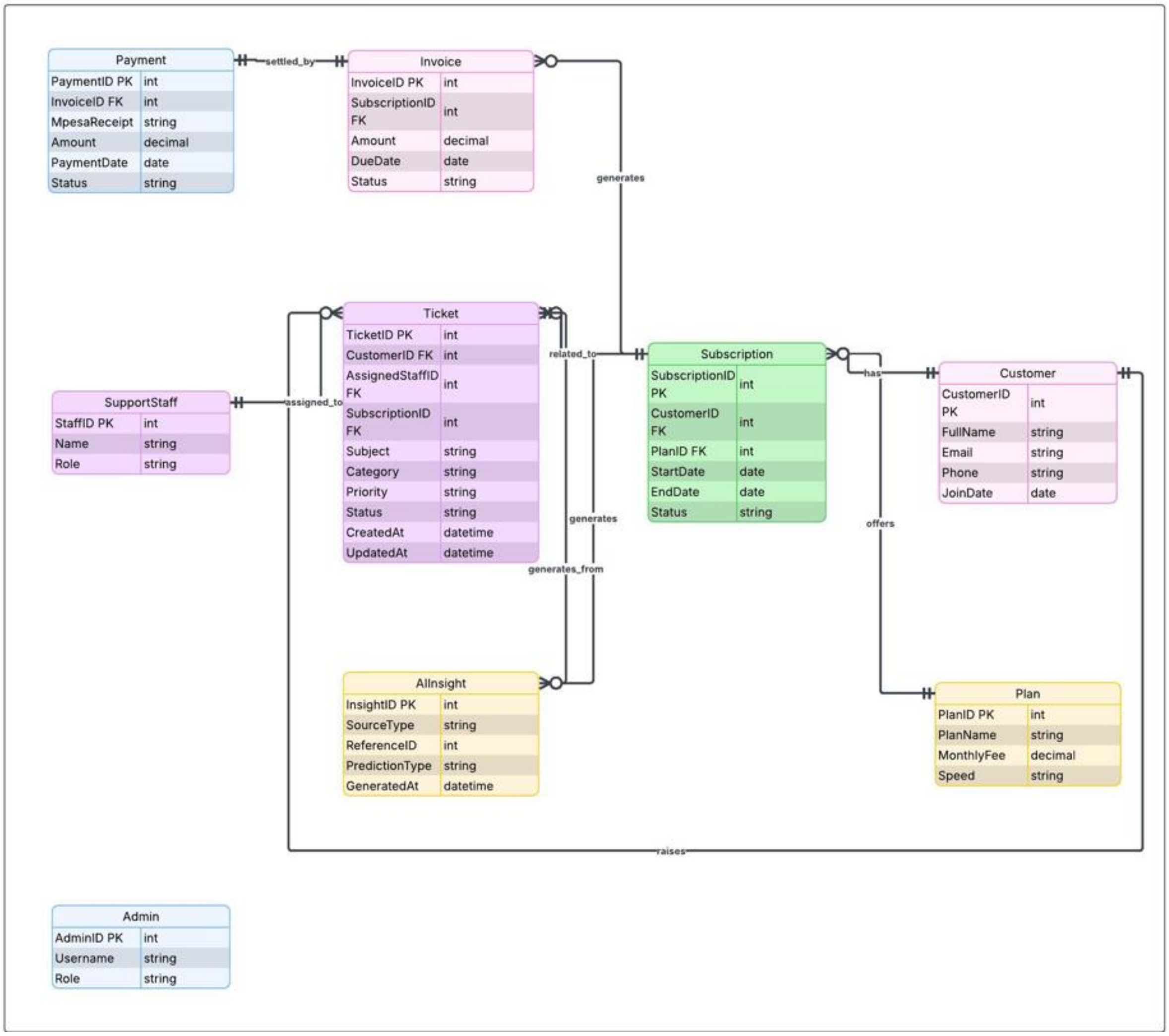


*Figure D4: The ERD shows the key database entities and their relationships in the ISP billing and management system.*

**Figure D.5 Data Flow Diagram**

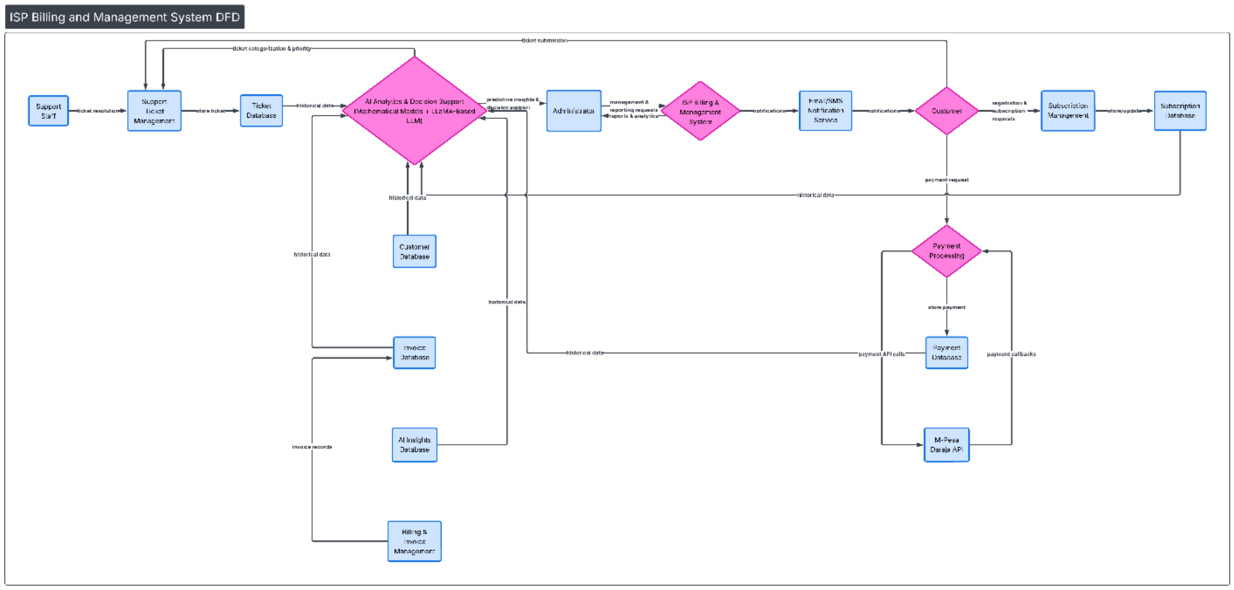


*Figure D5: The DFD illustrates the data flows in the ISP's billing and management system.*

These diagrams collectively provide a clear visual representation of the system's structure, behavior, and data organization, supporting both implementation and evaluation of the proposed solution.